

VERIFIX

(A Verified Artisan Hub)

Cybersecurity Design Assessment Report Database Schema & Architecture Review

Executive Summary

This assessment evaluates the security implications of the proposed VERIFIX database schema and system architecture before implementation begins. The review focuses on data storage, identity verification, payment processing, access control, and backend design decisions.

Assessment Scope

This assessment reviews the security considerations associated with the Users, Artisan Profiles, Jobs, Matches, and Transactions collections, as well as authentication, authorization, file uploads, payment processing, data protection, and monitoring requirements.

Security Findings & Recommendations

Finding	Risk	Observation	Recommendation
Identity Verification Data Protection	Critical	Sensitive ID documents stored.	Restrict access, private storage, audit access.
Role-Based Access Control	Critical	Admin roles not defined.	Implement RBAC with admin roles.
Payment Processing Security	Critical	Revenue-related transactions.	Validate webhooks and server-side payment logic.
Firestore Security Rules	Critical	Access rules not shown.	Apply deny-by-default access controls.
Input Validation	High	User input fields may be abused.	Validate and sanitize all inputs.
File Upload Security	High	Uploads may contain malicious files.	Restrict file types and sizes.
Authentication Security	High	OTP abuse possible.	Rate limit and expire OTPs.
Audit Logging	Medium	No audit trail visible.	Implement security event logging.

Conclusion

The VERIFIX design is a strong functional foundation but requires security controls around access control, payment integrity, Firestore security rules, and identity document protection before production use.